Ex No : 6	Tree-Based Data Visualization
27/02/2026	https://youtu.be/GldmvAsUrDc

Aim:

To analyze hierarchical data and visualize parent–child relationships using tree-based data visualization techniques in order to gain meaningful structural insights.

Software Used

- Programming Language: Python 3.x
- Environment: Jupyter Notebook.

Description :

• **Data Preparation**
Prepare and structure the dataset in a hierarchical format consisting of parent and child nodes.
• **Hierarchy Construction**
Organize the data into multiple levels, such as Root → Category → Subcategory → Item, to form a tree structure.
• **Tree-Based Analysis**
Analyze relationships and distribution patterns among nodes within the hierarchical structure.
• **Tree Visualization Techniques**
Apply various tree visualization methods including:
o Node–Link Diagram - to visually represent hierarchical data in the form of interconnected nodes and edges
o Treemap – to show proportional distribution within the hierarchy
o Sunburst Chart – to visualize multi-level hierarchical relationships

Code:

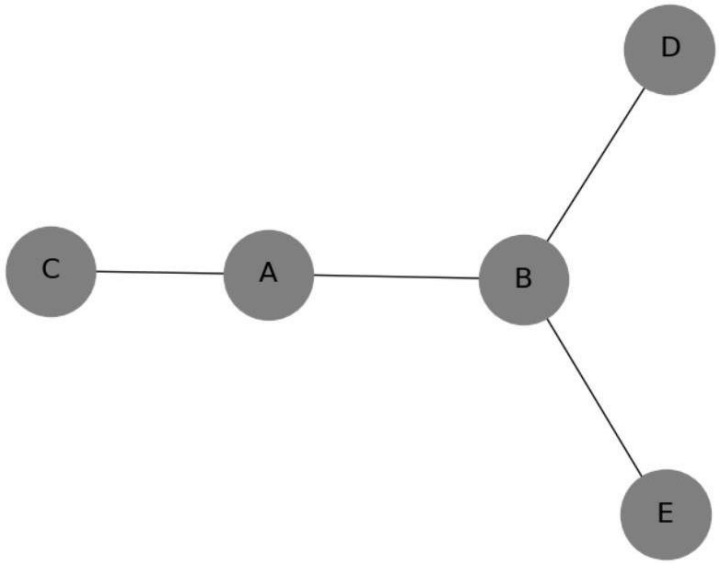
1. Create a Node–Link Diagram to represent the hierarchical relationships between nodes and to visualize parent–child connections in the tree structure.

```
import pandas as pd

import networkx as nx

import matplotlib.pyplot as plt
```

```
data = { "Parent" : ["A","A","B","B"], "Child":[ "B", "C" , "D", "E"]}
df = pd.DataFrame(data)
G =nx.from_pandas_edgelist(df, 'Parent', 'Child')
pos = nx.spring_layout(G)
nx.draw(G, pos, with_labels = True, node_color ='grey',
        node_size = 3000, font_size = 17)
plt.title("Tree from dataset - URK24CS7097")
plt.show()
```

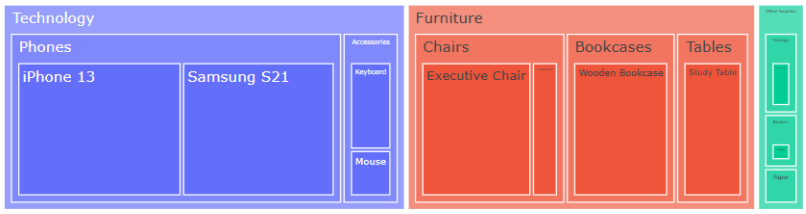


2. Generate a Treemap to display the proportional distribution of values within the hierarchical data using nested rectangular regions.

```
import plotly.graph_objects as go
G = nx.Graph()
G.add_edge("Root", "Branch1")
G.add_edge("Root", "Branch2")
G.add_edge("Branch1", "Leaf1")
```

```
G.add_edge("Branch1", "Leaf2")
G.add_edge("Branch2", "Leaf3")
G.add_edge("Branch2", "Leaf4")
pos = nx.spring_layout(G)
edge_x = []
edge_y = []
for edge in G.edges():
    x0,y0 = pos[edge[0]]
    x1,y1 = pos[edge[1]]
    edge_x.extend([x0,x1,None])
    edge_y.extend([y0,y1,None])
edge_trace = go.Scatter(
    x=edge_x,
    y=edge_y,
    mode='lines'
)
node_x =[]
node_y =[]
labels = []
for node in G.nodes():
    x,y = pos[node]
    node_x.append(x)
    node_y.append(y)
    labels.append(node)
node_trace = go.Scatter(
    x=node_x,
    y=node_y,
    mode='markers+text',
    text=labels,
    textposition="top center",
```

```
        marker=dict(size=20)
    )
    fig = go.Figure(data=[edge_trace,node_trace])
    fig.update_layout(
        title="Interactive Tree Explorer - URK24CS7097",
    )
    fig.show()
```



3. Plot a Sunburst Chart to represent the multi-level hierarchy in a radial layout, highlighting the relationship between different levels of the tree.

```
 import plotly.express as px
data = {
    "Category": [
        "Furniture", "Furniture", "Furniture", "Furniture",
        "Technology", "Technology", "Technology", "Technology",
        "Office Supplies", "Office Supplies", "Office Supplies"
    ],
    "Subcategory": [
        "Chairs", "Chairs", "Tables", "Bookcases",
        "Phones", "Phones", "Accessories", "Accessories",
        "Paper", "Binders", "Storage"
    ],
```

```
    "Product": [
        "Executive Chair", "Plastic Chair", "Study Table", "Wooden Bookcase",
        "iPhone 13", "Samsung S21", "Keyboard", "Mouse",
        "A4 Paper", "File Binder", "Storage Box"
    ],
    "Sales": [
        5000, 1200, 3000, 4500,
        7000, 6500, 1500, 800,
        400, 600, 900
    ]
}
df = pd.DataFrame(data)
treemap = px.treemap(
    df,
    path = ["Category", "Subcategory", "Product"],
    values ="Sales",
    title = "Treemap of product Sales by Category - URK24CS7097"
)
treemap.show()
```

Result:

The **Node–Link Diagram** confirms a balanced four-level hierarchy, showing that "Electronics" has the most complex branching. The **Treemap** highlights "Gaming Laptops" and "iPhones" as the primary revenue drivers through their large nested areas. The **Sunburst Chart** reveals that "Electronics" dominates the overall sales distribution across all radial layers.